Lecture 16 (10/21)

Leaving out predictors and the sample distribution

Last class:

- Omitted variables in AR processes
- Categorical predictors and dummy variables
- **Example:** Incumbency advantage in congressional elections

LAST DAY OF MIDTERM 2 MATERIAL

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- Omitted variables in AR processes
- Categorical predictors and dummy variables
- **Example:** Incumbency advantage in congressional elections

Today:

1) **Example:** earnings, elections, test scores
2) Leaving out all predictors but one
3) The sample distribution with correlated predictors

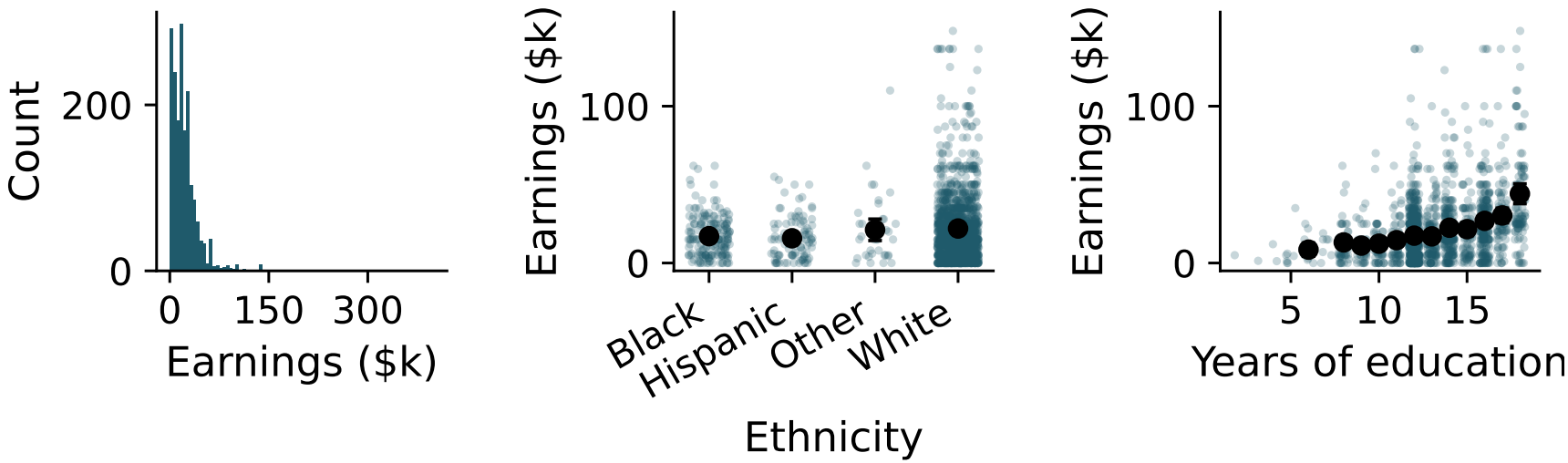
LAST DAY OF MIDTERM 2 MATERIAL

Example

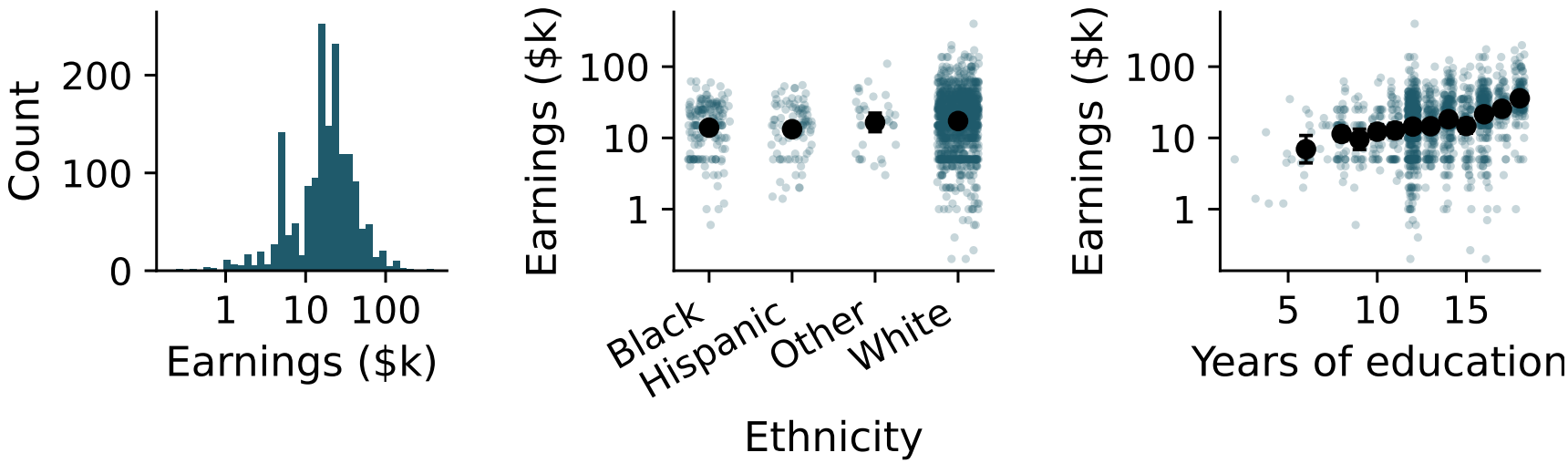
How are earnings related to gender and education? Survey of 1814 US adults (from *Regression and Other Stories*):

- Y (`earnk`): earnings in \$1000s;
- $X_{\text{hispanic}}, X_{\text{other}}, X_{\text{white}}$ (`hispanic`, `other`, `white`): dummy variables for ethnicity, with Black as the baseline;
- X_{male} (`male`), X_{height} (`height`, inches), X_{educ} (`education`, years), X_{age} (`age`, years).

Earnings vs. ethnicity and education



Linear scale. Note: 4 respondents earning more than $160k are not shown.



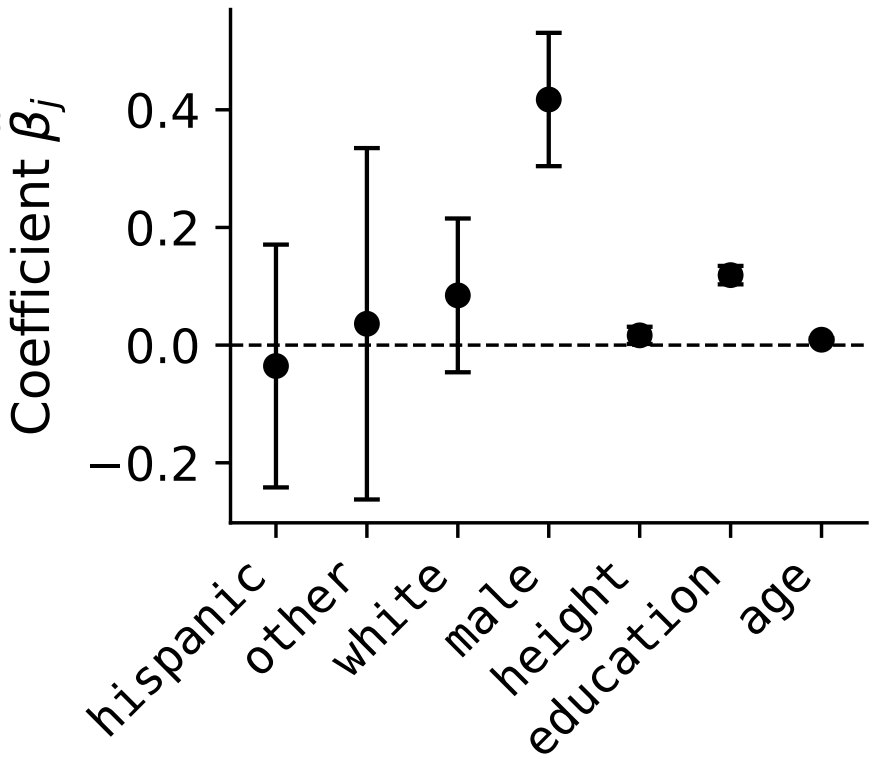
Log scale using 1627 respondents with positive earnings. Black dots: averages with 95% CIs.

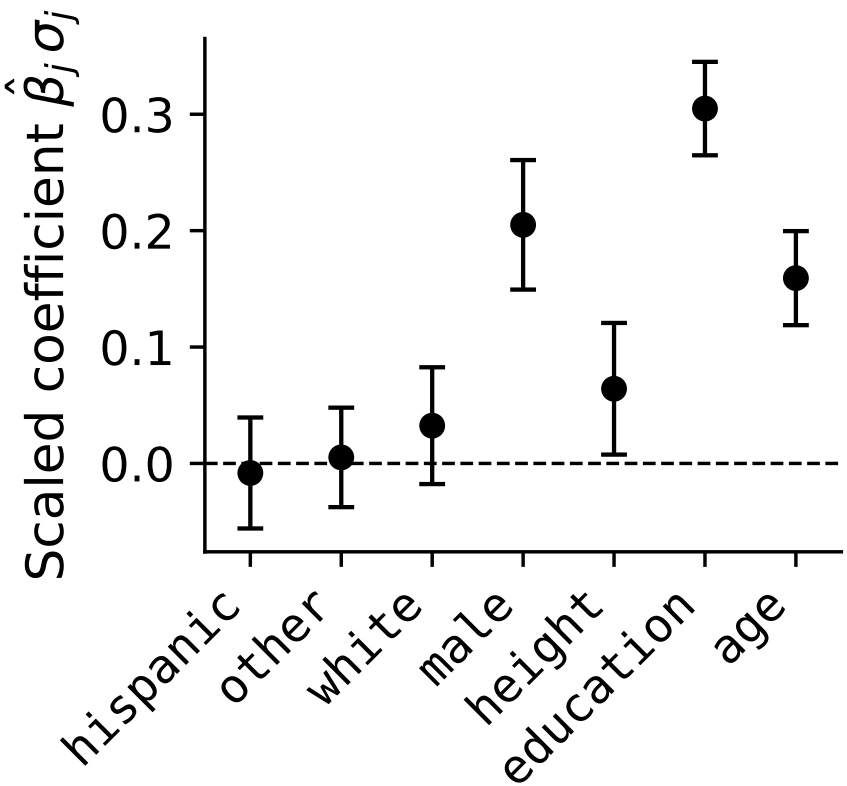
Earnings with all predictors

$$\log(\texttt{earnk}) = \beta_0 + \beta_{\text{hispanic}} X_{\text{hispanic}} + \beta_{\text{other}} X_{\text{other}} + \beta_{\text{white}} X_{\text{white}} + \beta_{\text{male}} X_{\text{male}} + \beta_{\text{height}} X_{\text{height}} + \beta_{\text{educ}} X_{\text{educ}} + \beta_{\text{age}} X_{\text{age}} + \epsilon$$

Dep. Variable:		log_earnk	R-squared:			0.211
	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.5156	0.496	-1.039	0.299	-1.489	0.458
hispanic	-0.0355	0.105	-0.337	0.736	-0.242	0.171
other	0.0363	0.152	0.238	0.812	-0.263	0.335
white	0.0845	0.067	1.268	0.205	-0.046	0.215
male	0.4175	0.058	7.222	0.000	0.304	0.531
height	0.0166	0.007	2.223	0.026	0.002	0.031
education	0.1189	0.008	14.885	0.000	0.103	0.135
age	0.0092	0.001	7.724	0.000	0.007	0.012

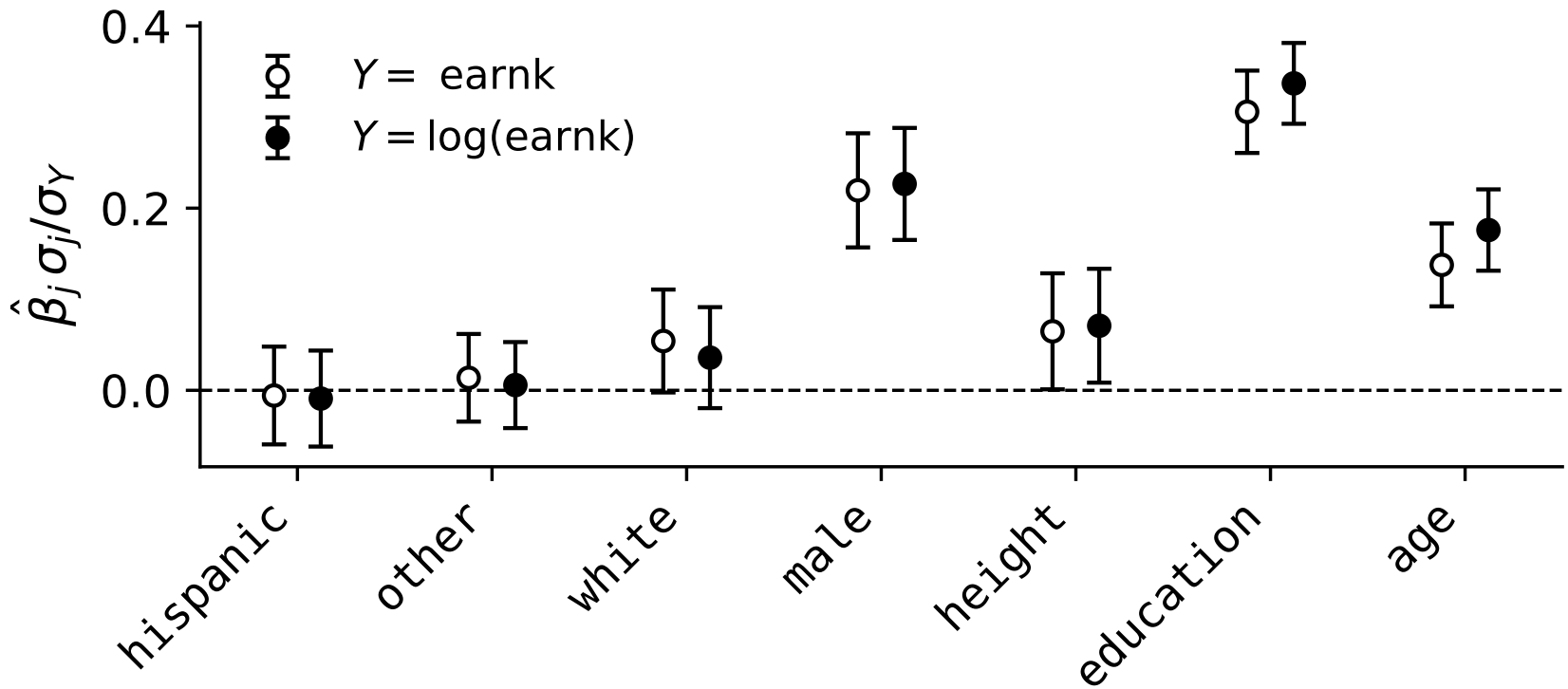
Coefficients vs. scaled coefficients



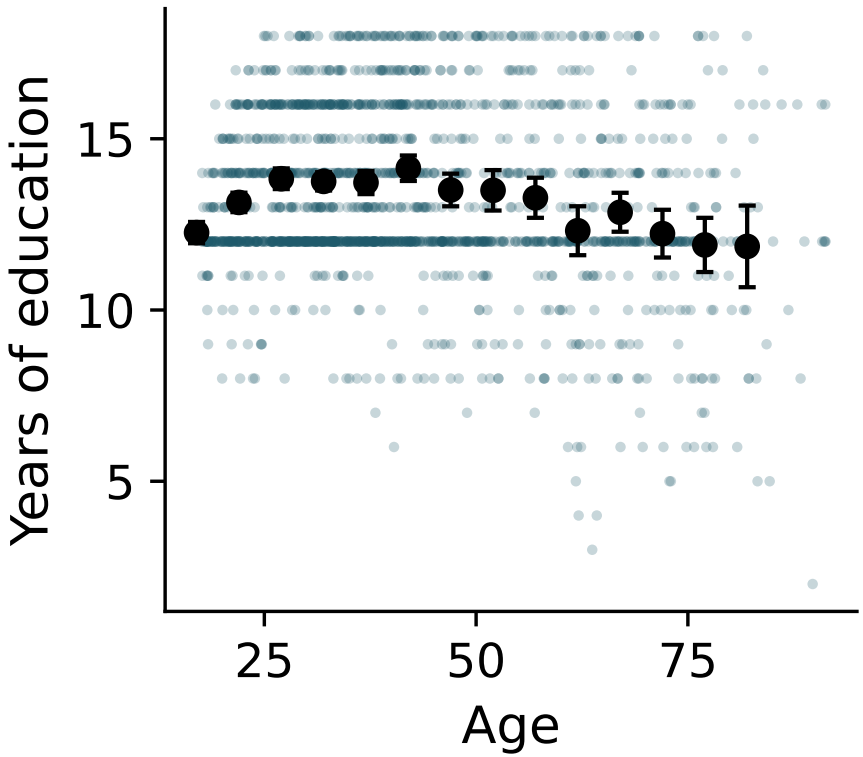


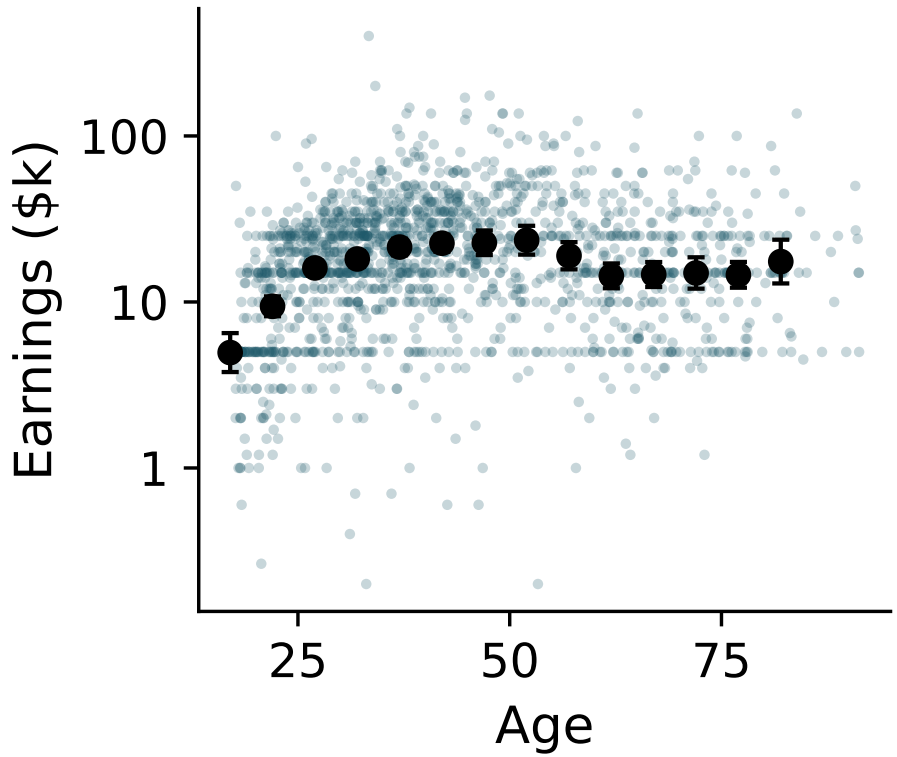
How important is the log transformation?

The full model fit to the 1627 respondents with positive earnings, with $Y = \texttt{earnk}$ ($R^2 = 0.18$) and with $Y = \log(\texttt{earnk})$ ($R^2 = 0.21$). Each coefficient is scaled by σ_j/σ_Y:

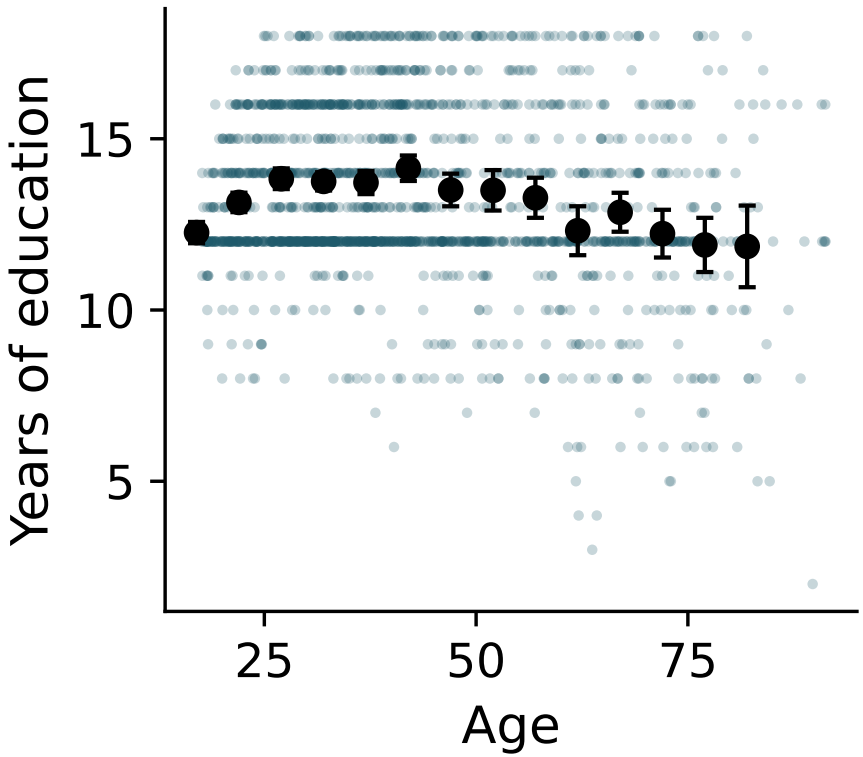


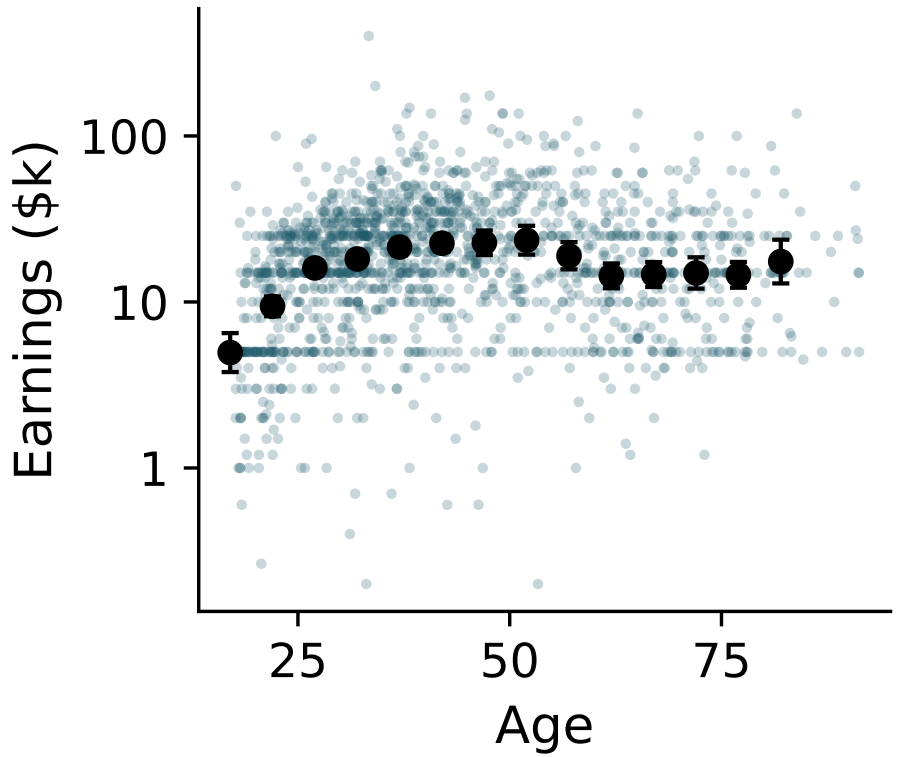
Education and earnings vs. age





Education and earnings vs. age

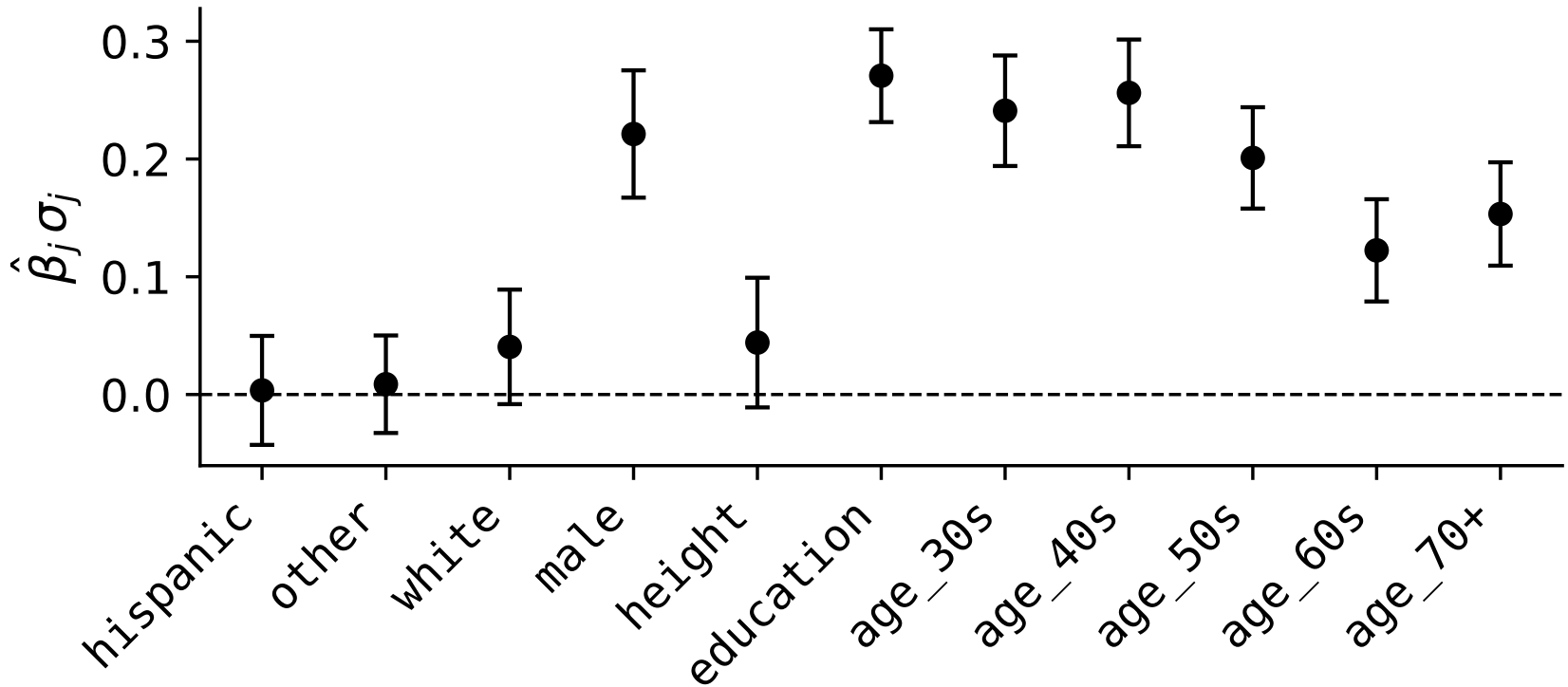




Explanation: Older respondents are from earlier generations, when less schooling was typical (the youngest are partly still in school).

Age as a categorical predictor

Replace age by one dummy variable per decade (baseline: 18–29, with 70 and over pooled), keeping the other predictors:



Omitting many predictors

What happens when we leave out many predictors?

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Theorem

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \epsilon$, *where* ϵ *is uncorrelated with* $X_1, \ldots, X_K$.

> *Omitted variable bias: if* $X_2, \ldots, X_K$ *are left out, the slope of the regression of* Y *on* X_1 *is*

$$\boxed{\beta_1' = \beta_1 + \sum_{j=2}^{K} \beta_j \beta_{1,j}'}, \qquad \beta_{1,j}' = \frac{cov(X_1, X_j)}{var(X_1)}.$$

Omitting many predictors

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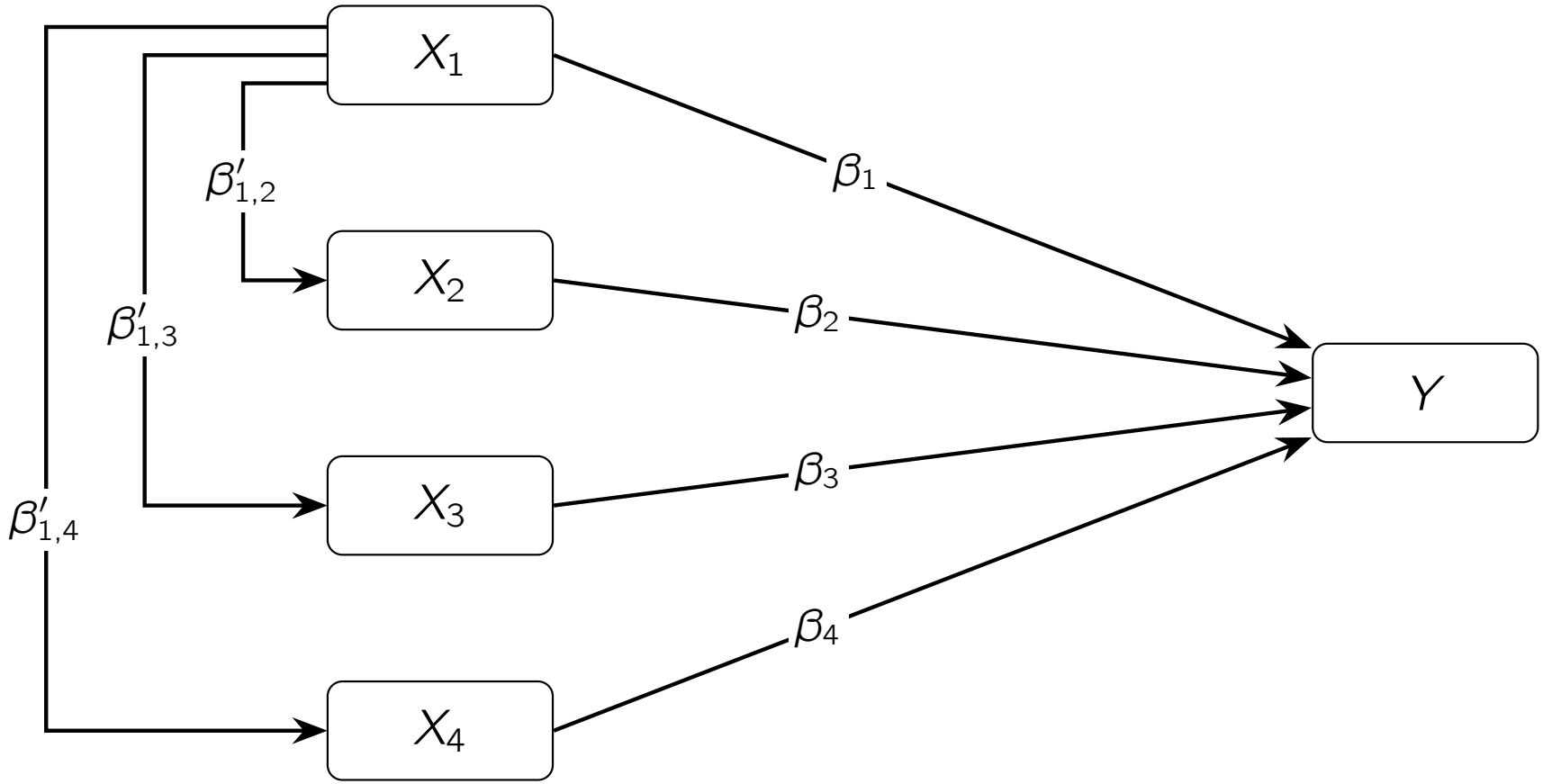
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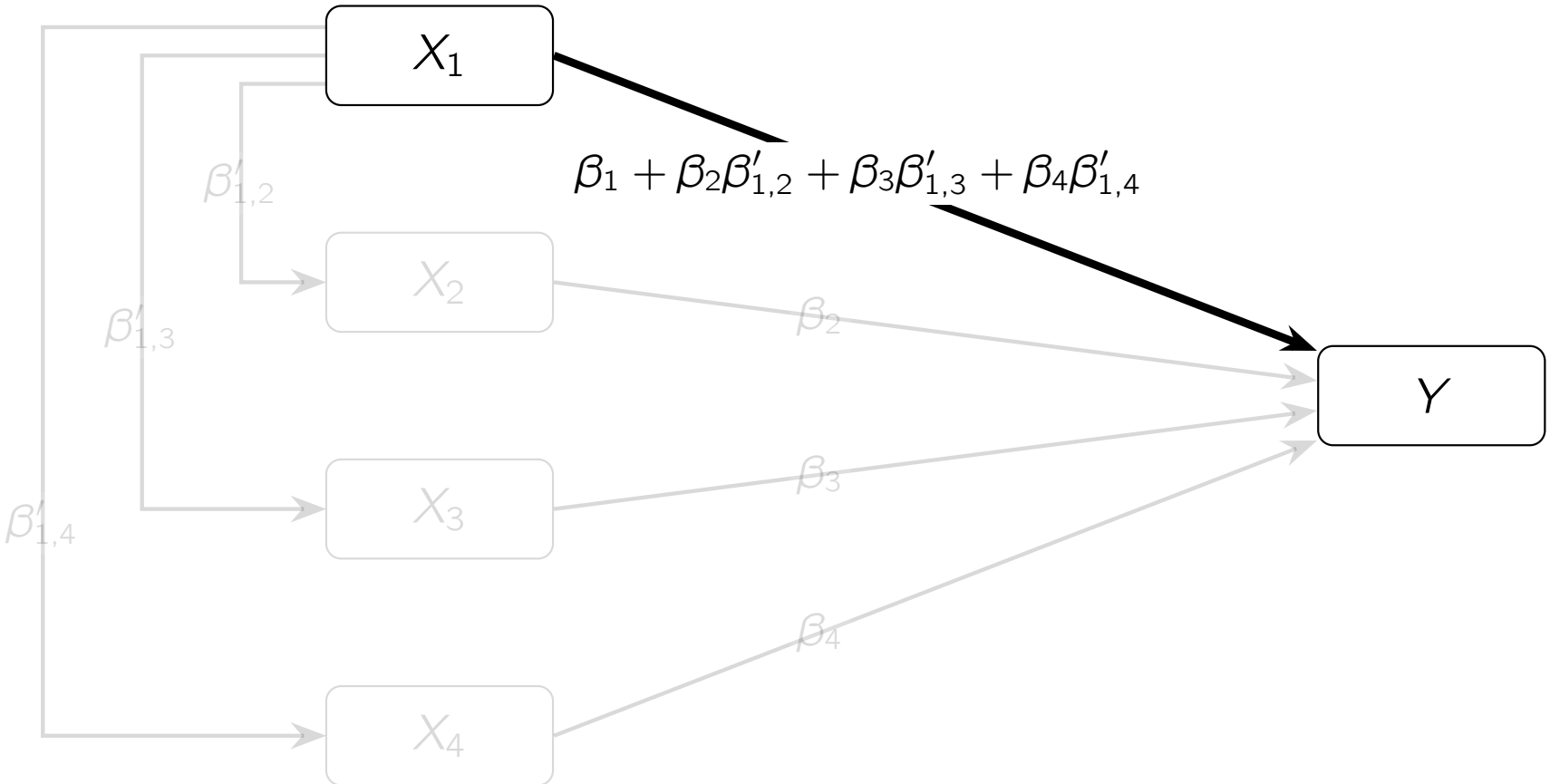
Question

Derive this formula, assuming $E[X_j] = 0$ for $j = 1, \ldots, K$ and $\beta_0 = 0$ (hence $E[Y] = 0$).

Leaving out all predictors but one

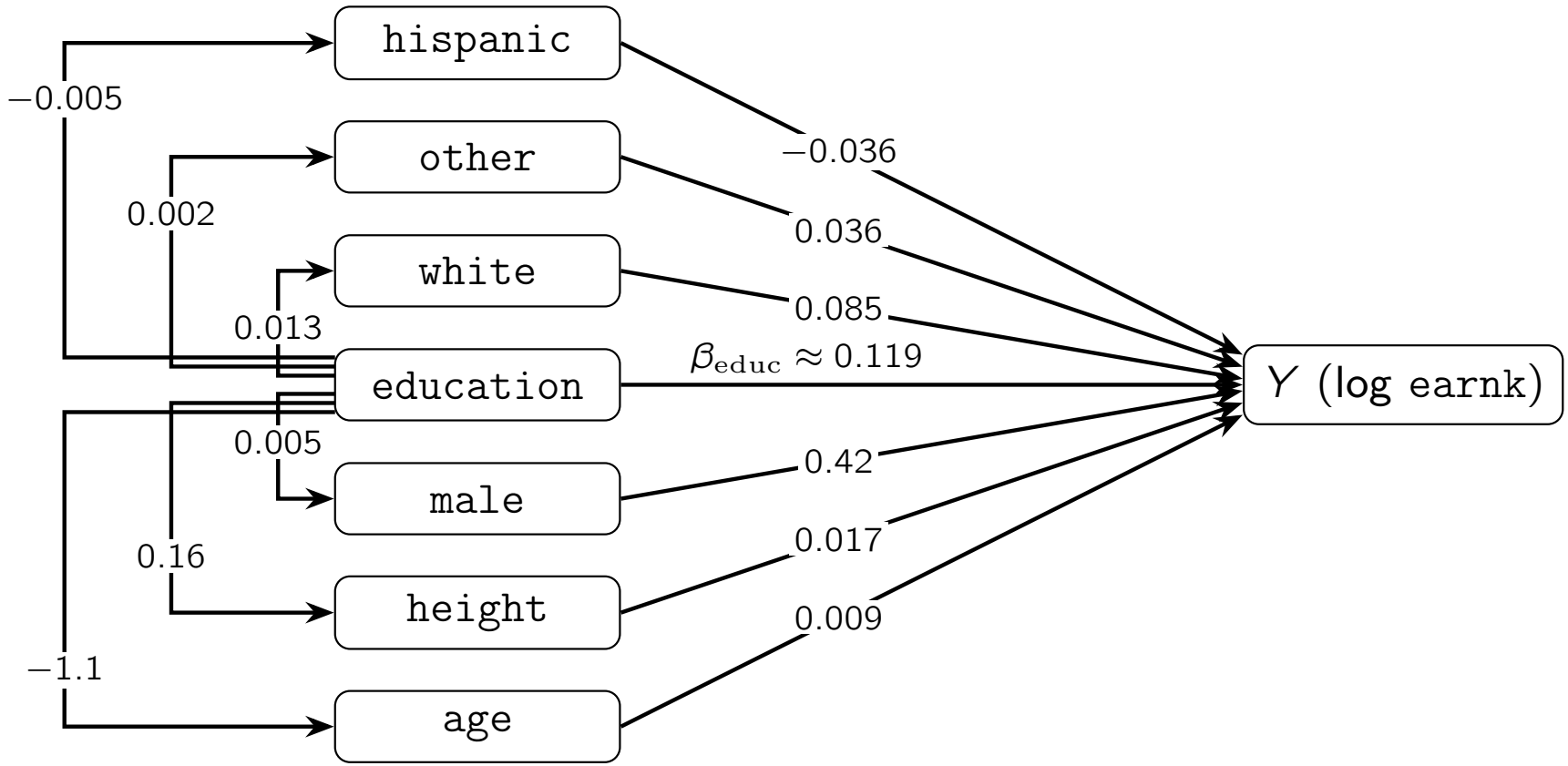


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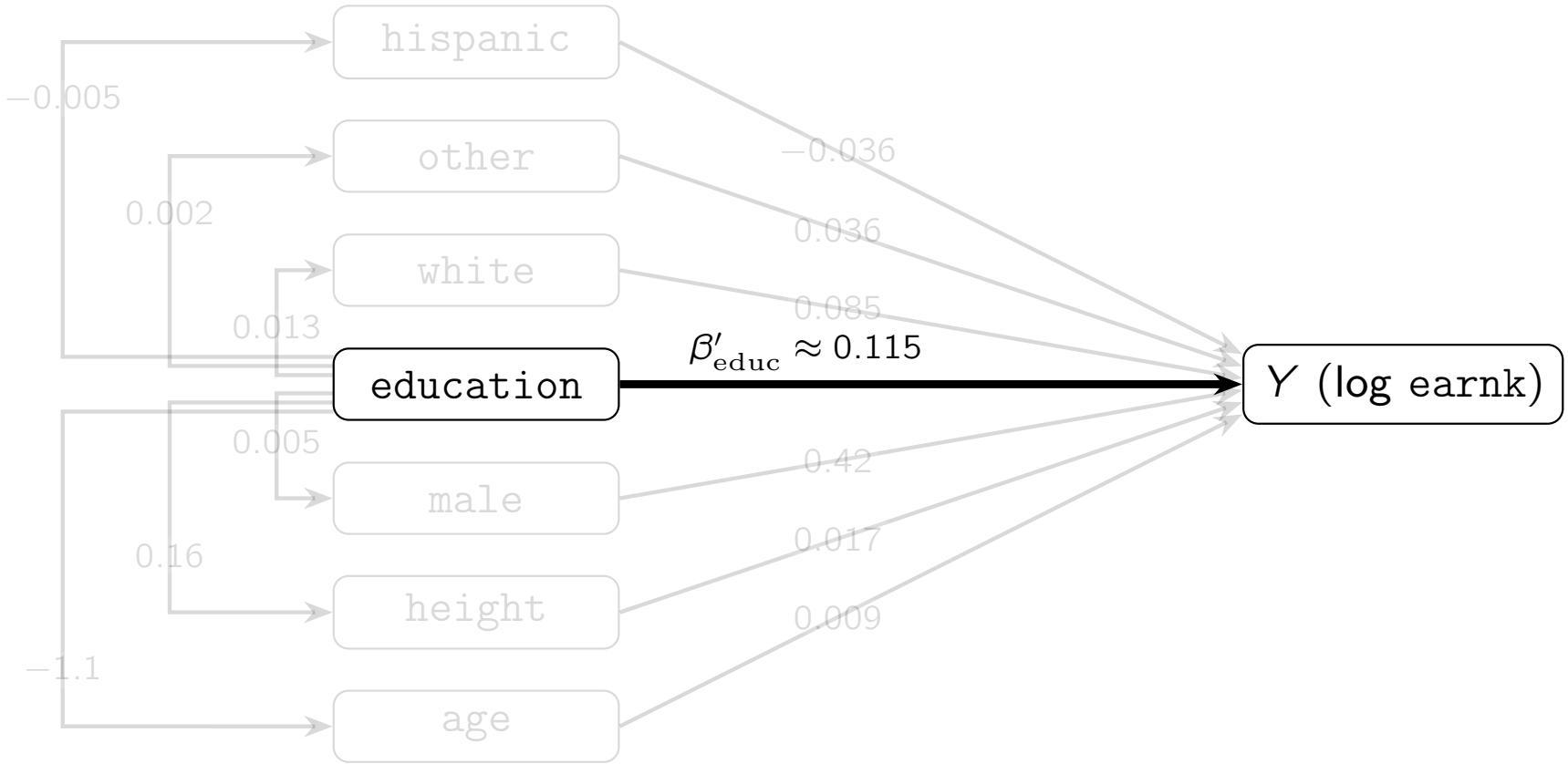
Here $\beta'_{1,j}$ is the single-predictor slope of X_j on X_1: this is the theorem for K predictors.

Leaving out all predictors but education



Left arrows: single-predictor slopes $\beta'_{\text{educ},j}$ of each predictor on `education`. Right arrows: coefficients β_j in the full model.

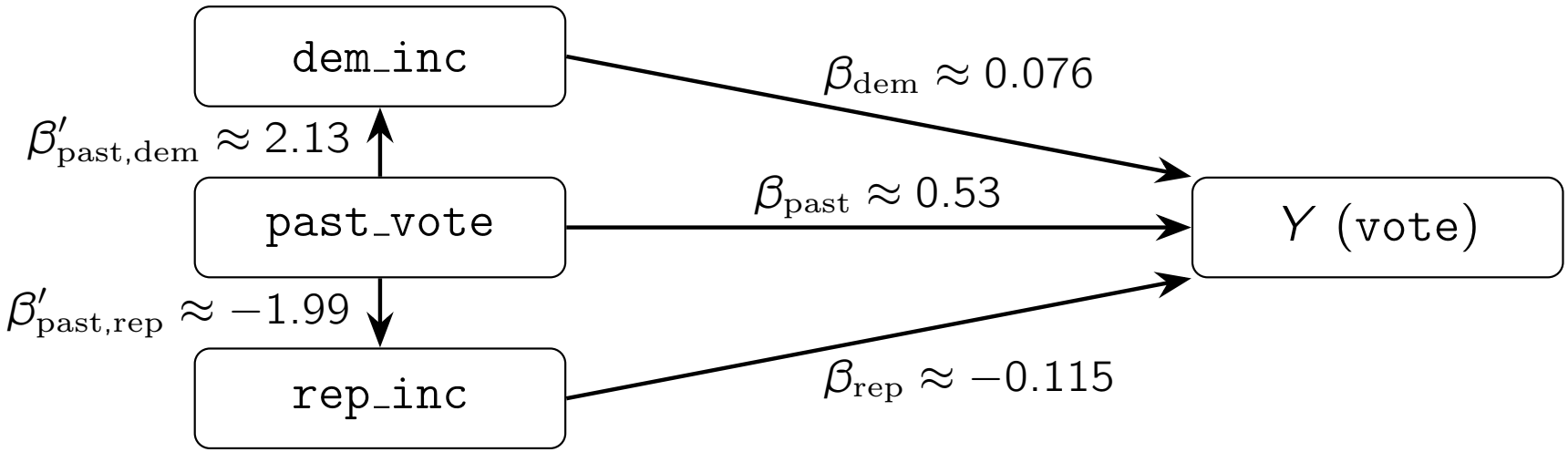
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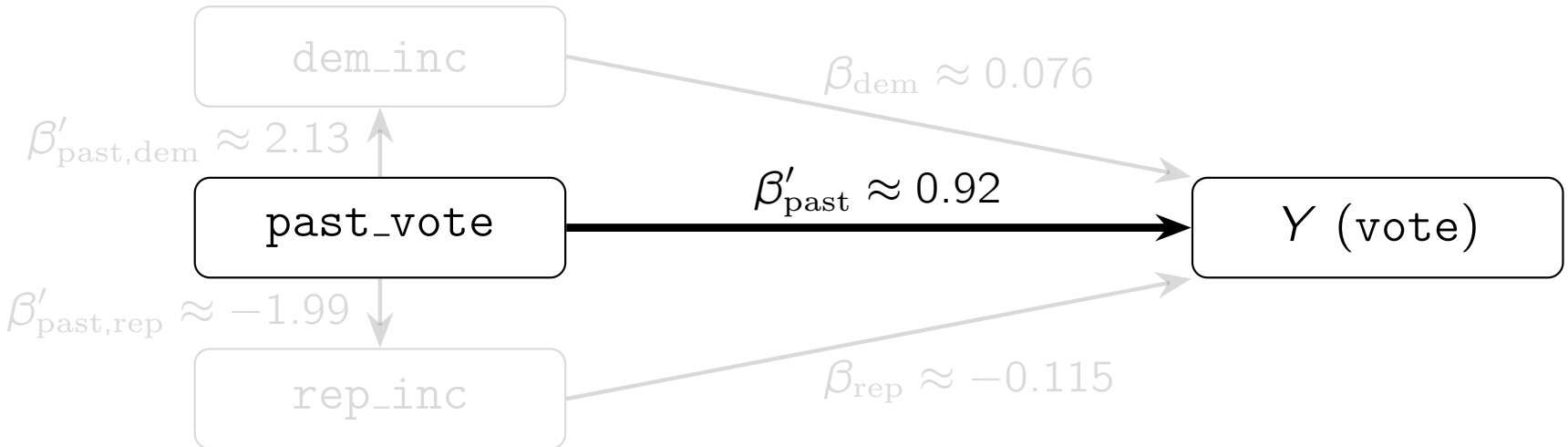
Left arrows: single-predictor slopes $\beta'_{\text{educ},j}$ of each predictor on `education`. Right arrows: coefficients β_j in the full model.

$$\underbrace{0.115}_{\beta'_{\text{educ}}} \approx \underbrace{0.119}_{\beta_{\text{educ}}} + \underbrace{0.001}_{\text{ethnicity (3 dummies)}} + \underbrace{0.42 \times 0.005}_{\text{male} \approx 0.002} + \underbrace{0.017 \times 0.16}_{\text{height} \approx 0.003} + \underbrace{0.009 \times (-1.1)}_{\text{age} \approx -0.010}$$

Revisiting elections: comparison to past vote only



Revisiting elections: comparison to past vote only



$$\underbrace{0.92}_{\beta'_{\text{past}}} \approx \underbrace{0.53}_{\beta_{\text{past}}} + \underbrace{0.076 \times 2.13}_{\beta_{\text{dem}}\beta'_{\text{past,dem}} \approx 0.16} + \underbrace{(-0.115) \times (-1.99)}_{\beta_{\text{rep}}\beta'_{\text{past,rep}} \approx 0.23}$$

Districts with a high past vote tend to have Democratic incumbents, and incumbency raises the vote.

Revisiting children's test scores: mother age

Children's test scores again, now adding X_{age} (mom_age), the mother's age at birth.

Dep. Variable:		kid_score	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	70.9569	8.307	8.542	0.000	54.631	87.283
mom_age	0.6952	0.362	1.920	0.055	-0.016	1.407

Dep. Variable:		kid_score	R-squared:			0.215
	coef	std err	t	P>|t|	[0.025	0.975]
const	20.9847	9.130	2.298	0.022	3.039	38.930
mom_hs	5.6472	2.258	2.501	0.013	1.210	10.085
mom_iq	0.5625	0.061	9.276	0.000	0.443	0.682
mom_age	0.2248	0.331	0.680	0.497	-0.425	0.875

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Question

Interpret the coefficient on `mom_age` in each model. With only `mom_hs` and `mom_iq`, the coefficients were 5.95 and 0.564 and $R^2 = 0.214$. What does this tell us about association between age and the other predictors?

Mother's age vs. schooling and IQ

Dep. Variable:		mom_hs	R-squared:			0.046
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0423	0.164	0.258	0.797	-0.280	0.365
mom_age	0.0326	0.007	4.565	0.000	0.019	0.047
Dep. Variable:		mom_iq	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	88.4081	6.105	14.482	0.000	76.409	100.407
mom_age	0.5087	0.266	1.912	0.057	-0.014	1.032

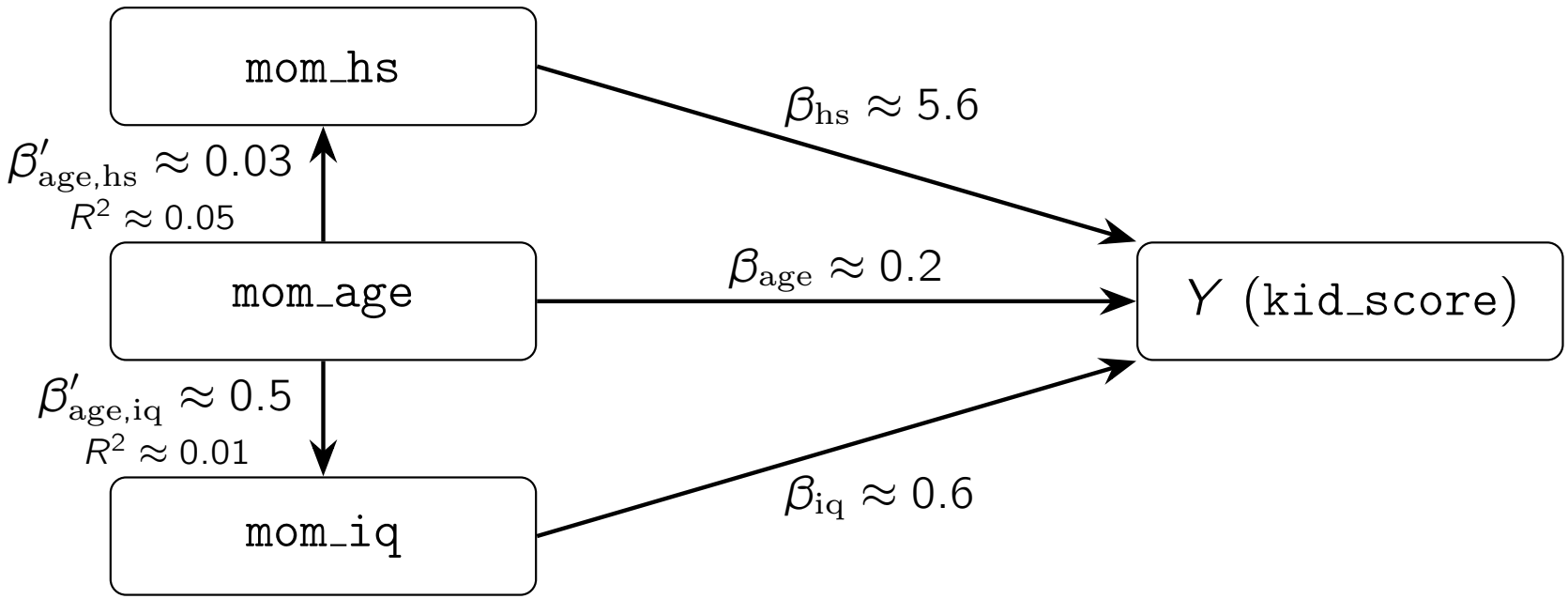
Mother's age vs. schooling and IQ

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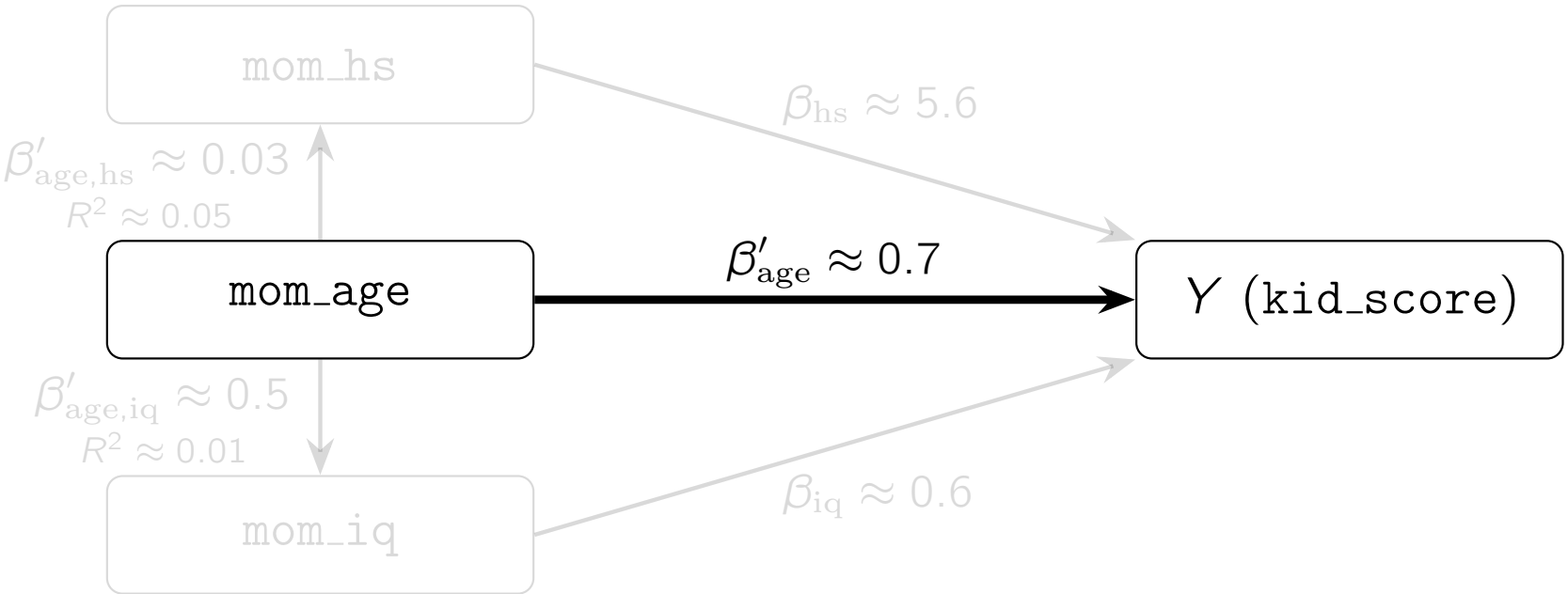
Question

Use the omitted variable bias formula to write the slope $\beta'_{\text{age}} \approx 0.70$ of the single-predictor regression of Y on X_{age} in terms of the coefficients of the three-predictor model and the slopes $\beta'_{\text{age,hs}}$ and $\beta'_{\text{age,iq}}$ above.

How mother's age is related to test scores



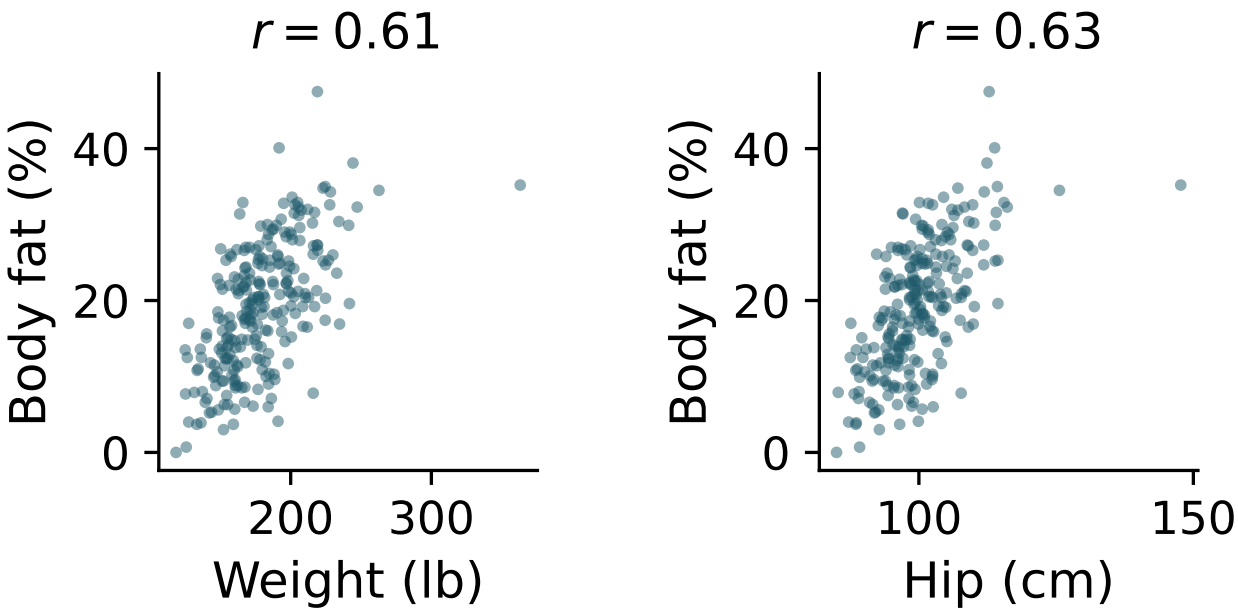
How mother's age is related to test scores



$$\underbrace{0.7}_{\beta'_{\text{age}}} \approx \underbrace{0.2}_{\beta_{\text{age}}} + \underbrace{0.03 \times 5.6}_{\beta'_{\text{age,hs}}\beta_{\text{hs}} \,\approx\, 0.2} + \underbrace{0.5 \times 0.6}_{\beta'_{\text{age,iq}}\beta_{\text{iq}} \,=\, 0.3}$$

Body fat, weight and hip size

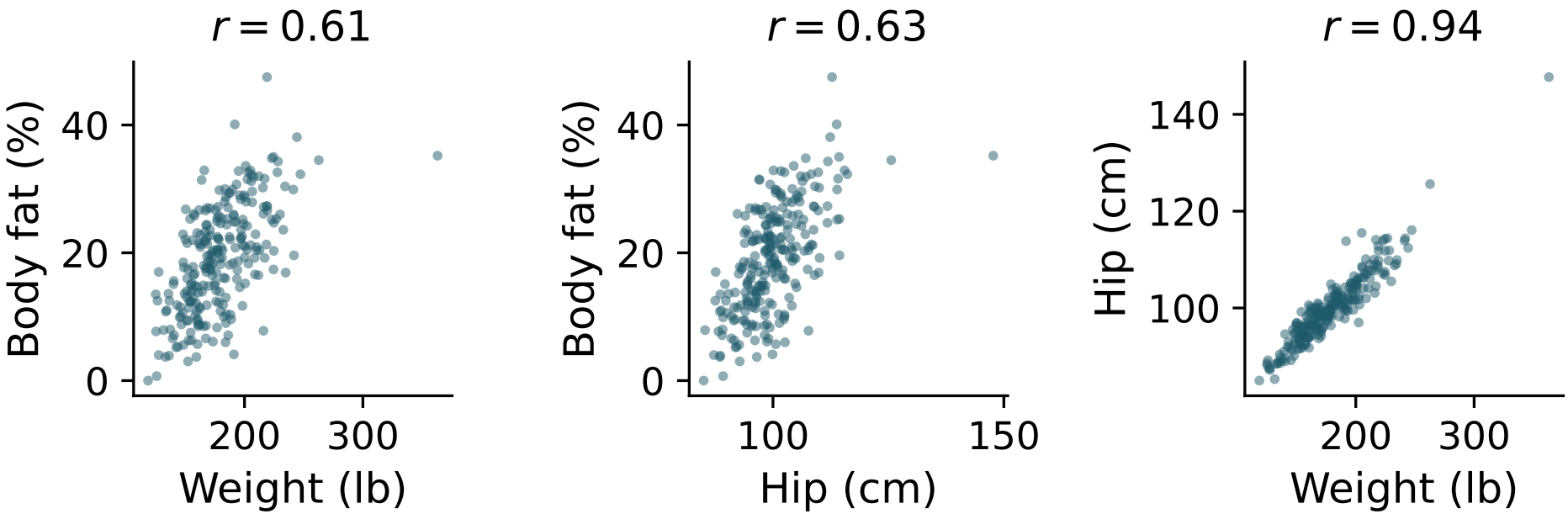
Percent body fat (from underwater weighing) of 252 men, with their weight (lb) and hip circumference (cm):



Data: Johnson (1996), *Journal of Statistics Education*.

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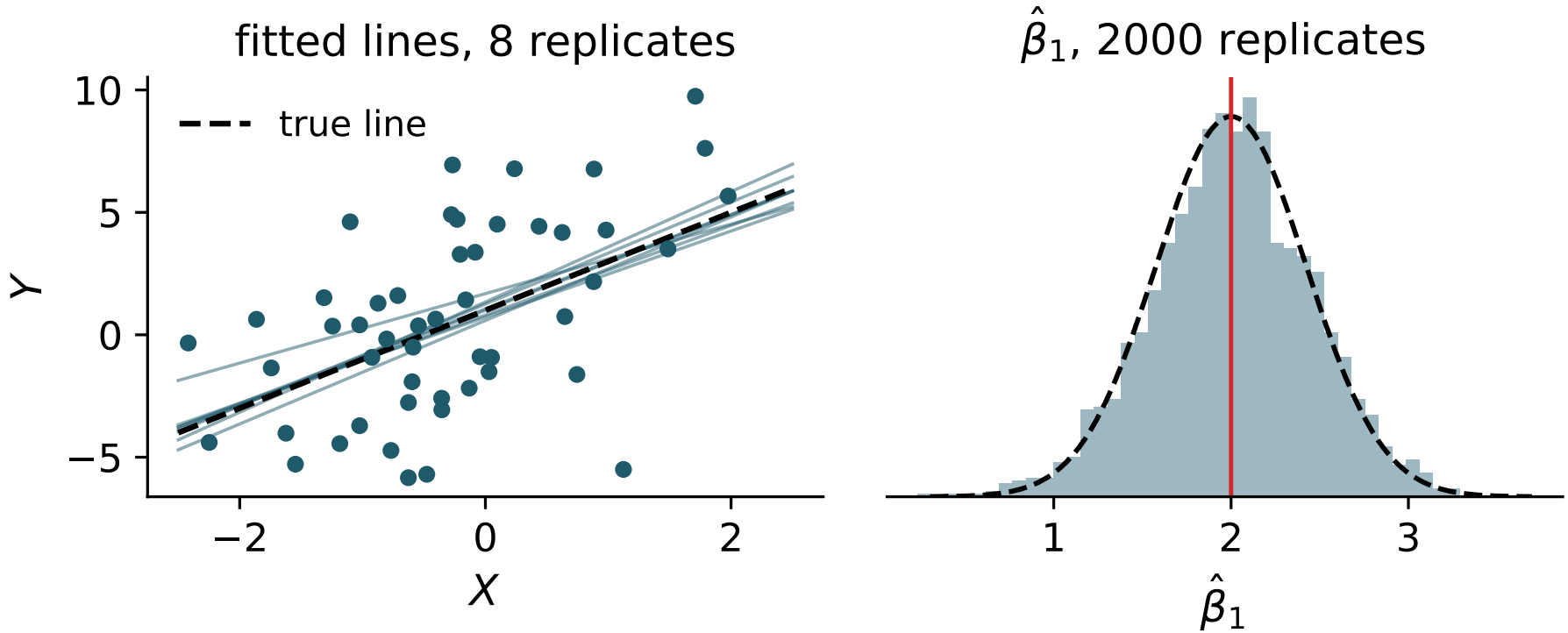
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Review: the sample distribution

Each replicate dataset of $N = 50$ is sampled from a linear regression model and yields different $\hat{\beta}_0$ and $\hat{\beta}_1$. The sample distribution is the distribution of these parameters over the replicates.

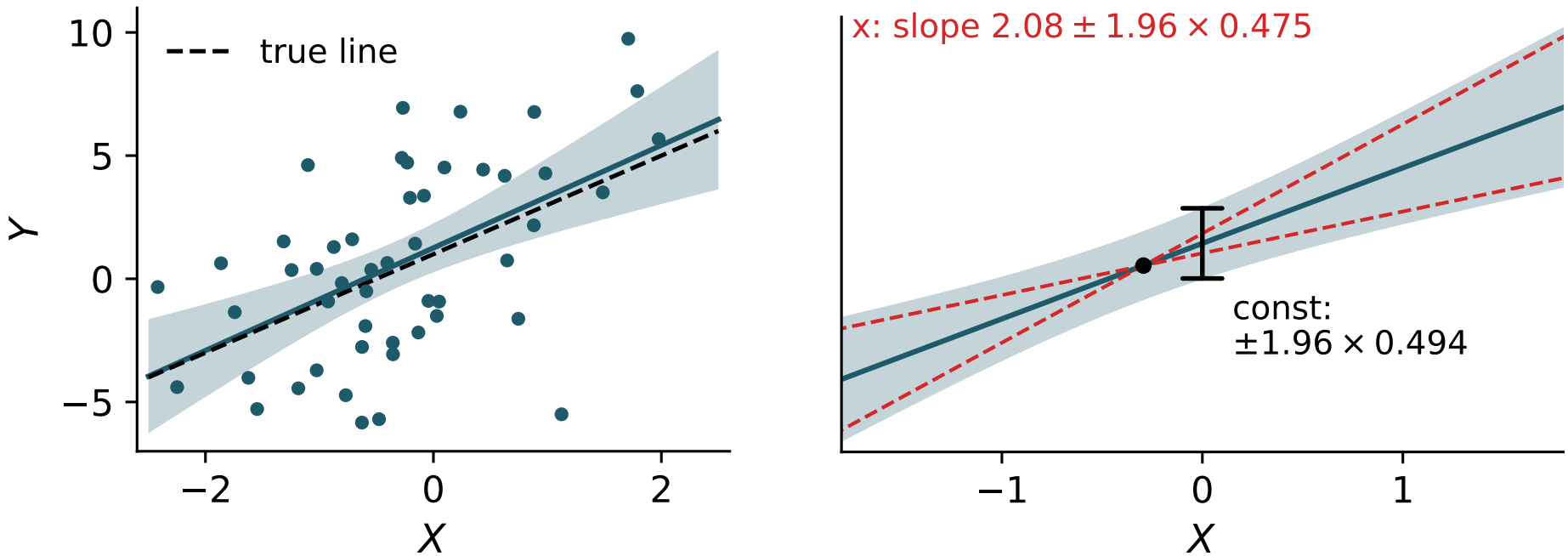


Dashed: Normal$\left(\beta_1,\ \sigma_\epsilon^2/(N\sigma_X^2)\right)$, so se$(\hat{\beta}_1) \approx \sigma_\epsilon/(\sigma_X\sqrt{N}) = 0.42$. Red: the true $\beta_1 = 2$. Remember that in practice we see only one dataset so se$(\hat{\beta}_1)$ must be estimated.

Review: one dataset

Output for the dataset shown on the previous slide:

Dep. Variable:			y	R-squared:		0.286
	coef	std err	t	P>|t|	[0.025	0.975]
const	1.2544	0.494	2.539	0.014	0.261	2.248
x	2.0832	0.475	4.389	0.000	1.129	3.038



Shaded: 95% confidence interval for the line. Right: at $X = 0$ the band is $\pm 1.96 \times 0.494$ (const); away from $\bar{x}$ (dot) it follows the slopes $2.08 \pm 1.96 \times 0.475$ (x).

Body fat: one predictor at a time vs. both

Dep. Variable:		bodyfat	R-squared:			0.375
	coef	std err	t	P>|t|	[0.025	0.975]
const	-12.0516	2.581	-4.669	0.000	-17.136	-6.968
weight	0.1744	0.014	12.249	0.000	0.146	0.202

Dep. Variable:		bodyfat	R-squared:			0.391
	coef	std err	t	P>|t|	[0.025	0.975]
const	-53.8129	5.775	-9.318	0.000	-65.187	-42.438
hip	0.7303	0.058	12.666	0.000	0.617	0.844

Dep. Variable:		bodyfat	R-squared:			0.396
	coef	std err	t	P>|t|	[0.025	0.975]
const	-41.4134	10.317	-4.014	0.000	-61.734	-21.093
weight	0.0600	0.041	1.449	0.149	-0.022	0.142
hip	0.4988	0.170	2.936	0.004	0.164	0.833

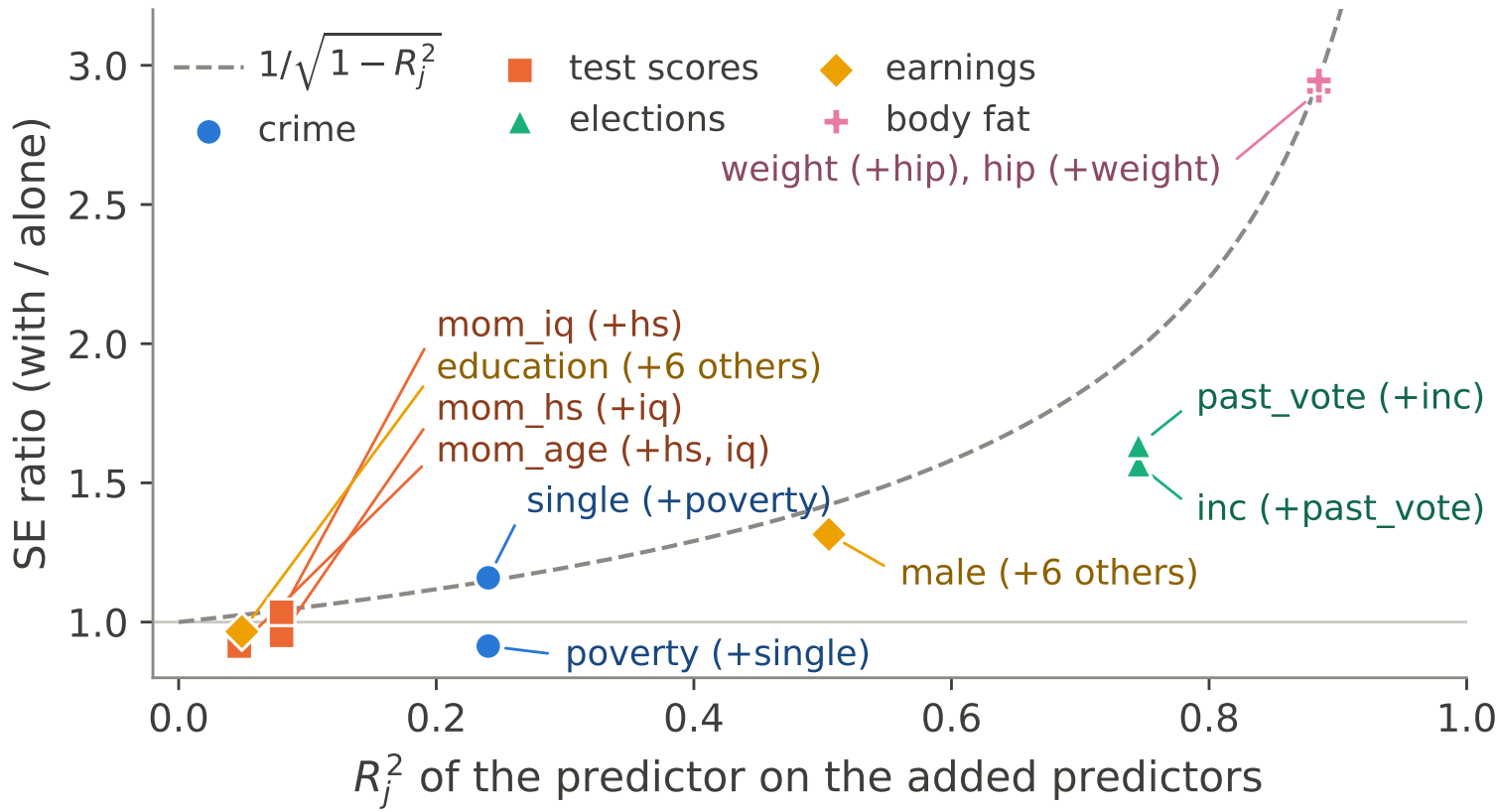
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hip	0.4988	0.170	2.936	0.004	0.164	0.833

Question

On its own, each predictor is strongly related to body fat. Why does the standard error of the weight coefficient triple, and weight stop being significant, when both are in the model?

Adding predictors and the standard error



The standard error grows when the added predictors are correlated with X_j (body fat, elections); when they are not, it barely changes.

The sample distribution with two predictors

Model with two standardized predictors with correlation $\rho_{1,2}$:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

With one predictor, $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / (\sigma_X \sqrt{N})$. With two, we need the *joint* sample distribution of $(\hat{\beta}_1, \hat{\beta}_2)$.

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It is helpful to understand the extreme case $X_1 \approx X_2$:

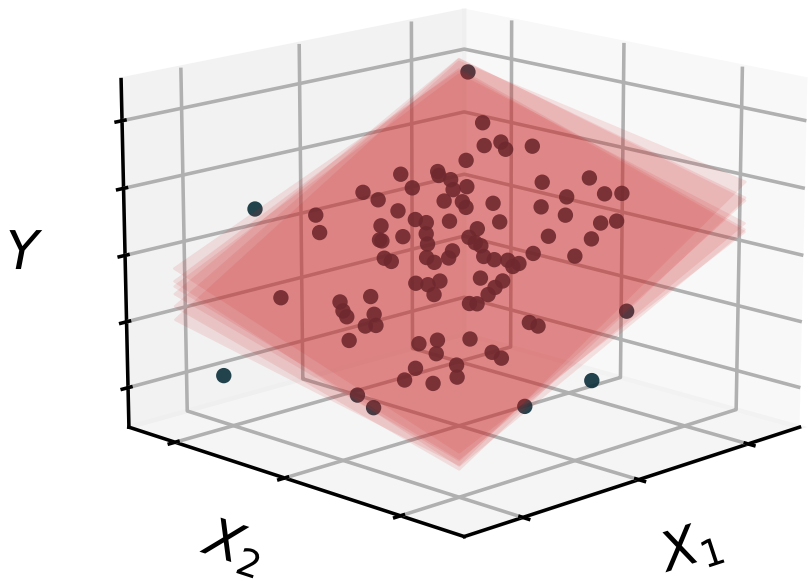
$$Y \approx (\beta_1 + \beta_2) X_1 + \epsilon.$$

The data pin down $\beta_1 + \beta_2$, but not β_1 and β_2 separately: raising $\hat{\beta}_1$ and lowering $\hat{\beta}_2$ by the same amount fits almost equally well.

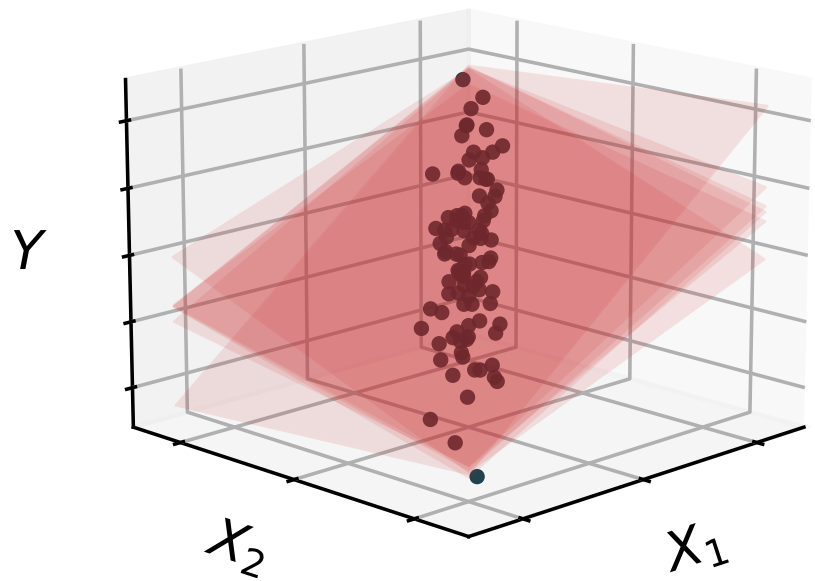
Geometric picture

Fitted planes from 6 datasets samples from the same model.

$\rho_{1,2} = 0$



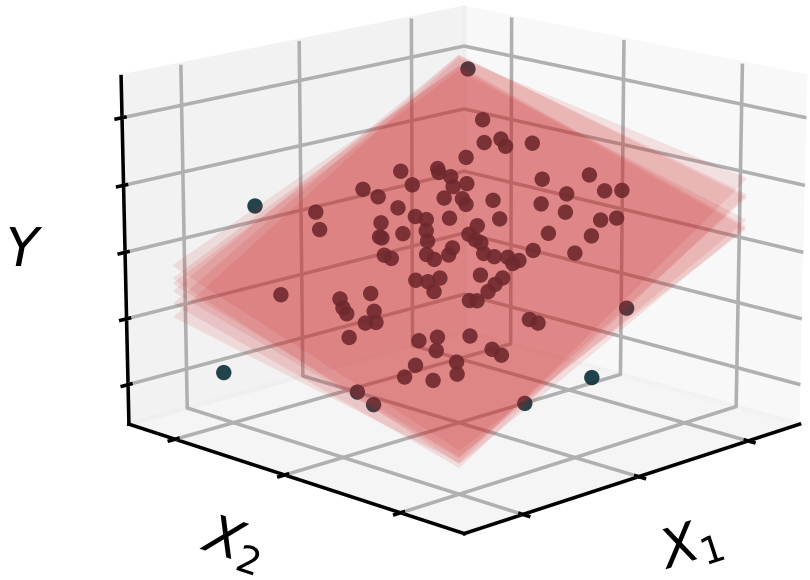
$\rho_{1,2} = 0.95$



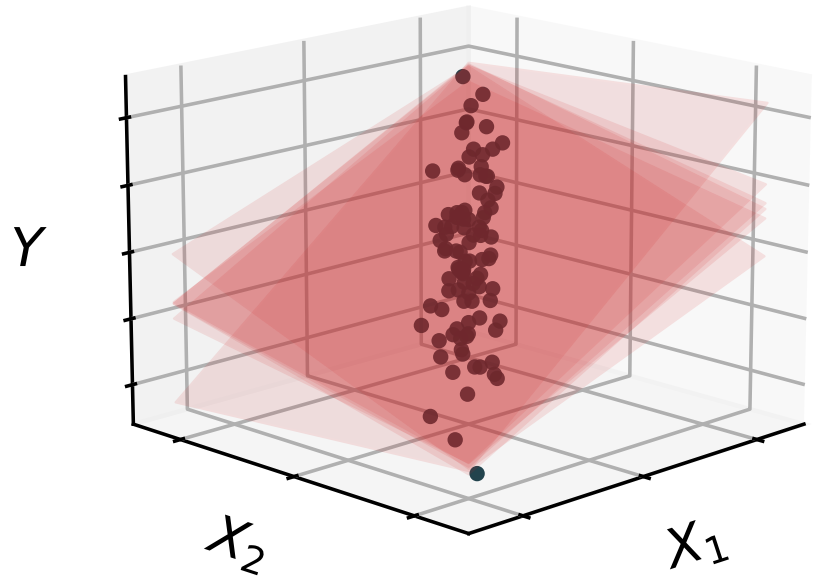
Geometric picture

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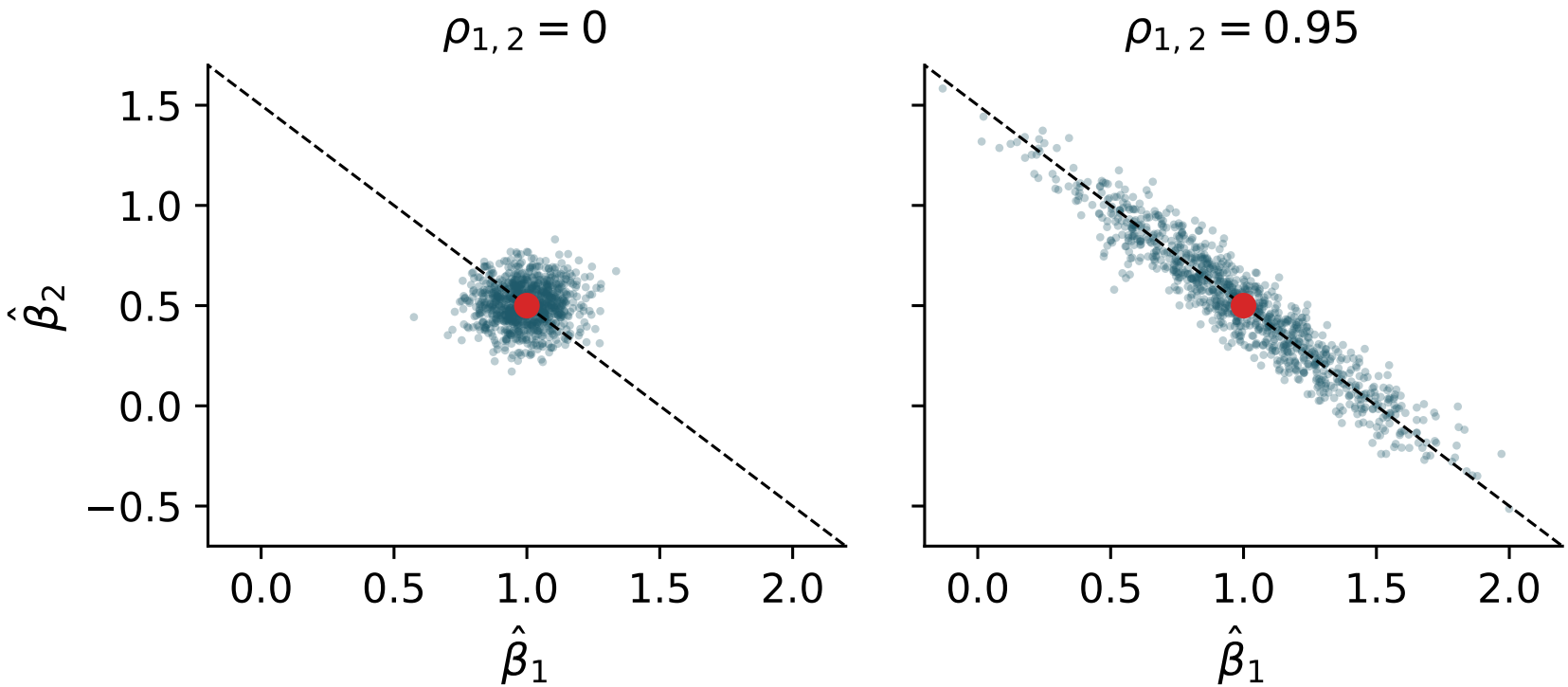
$\rho_{1,2} = 0.95$



With $X_1 \approx X_2$ the points lie near a line. The plane can tilt about that line without changing the fit much, so the individual slopes are poorly determined.

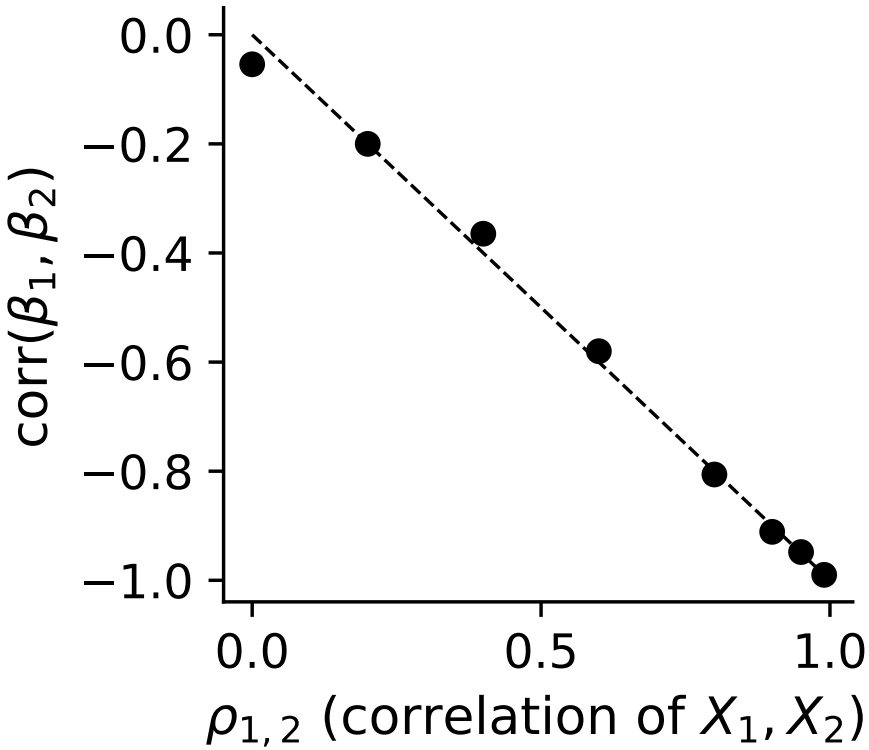
Simulated sample distribution

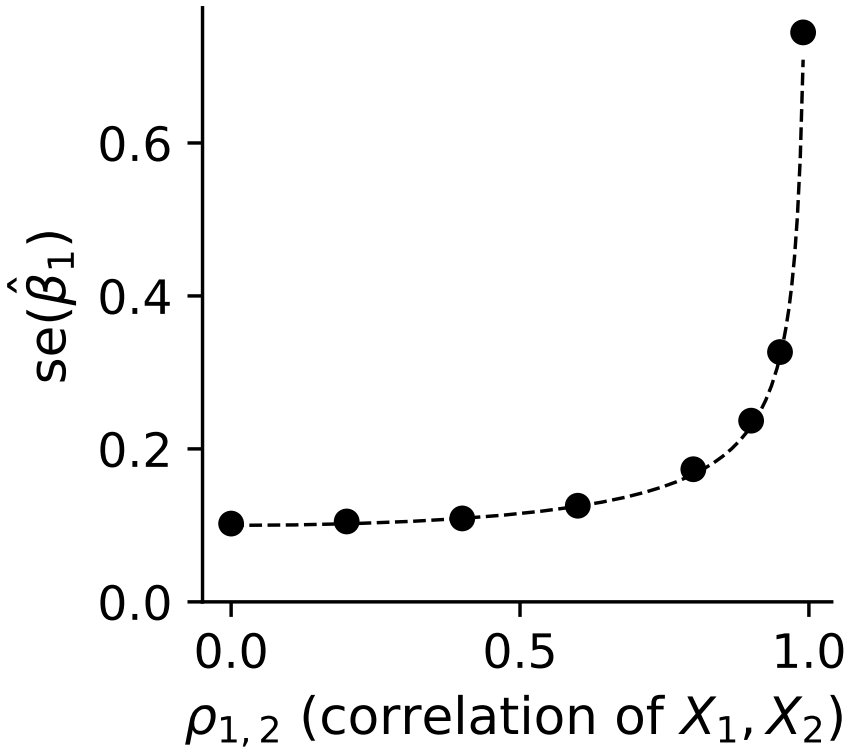
1000 datasets of $N = 100$ points from $Y = X_1 + 0.5X_2 + \epsilon$ ($\sigma_\epsilon = 1$):



Red: the true (β_1, β_2). Dashed: $\hat{\beta}_1 + \hat{\beta}_2 = 1.5$.

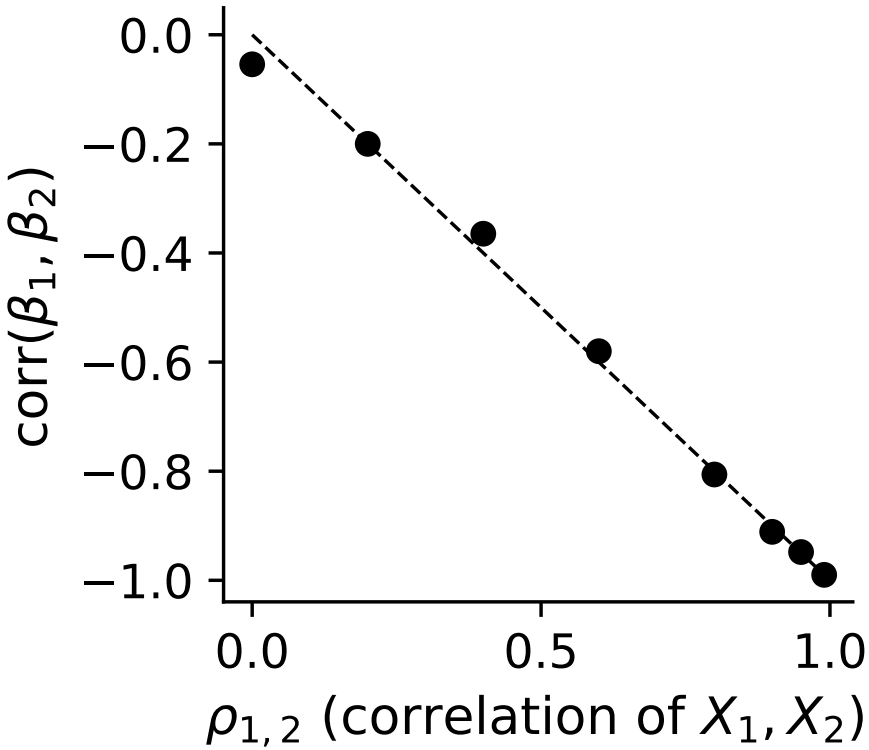
Correlation and standard error vs. $\rho_{1,2}$

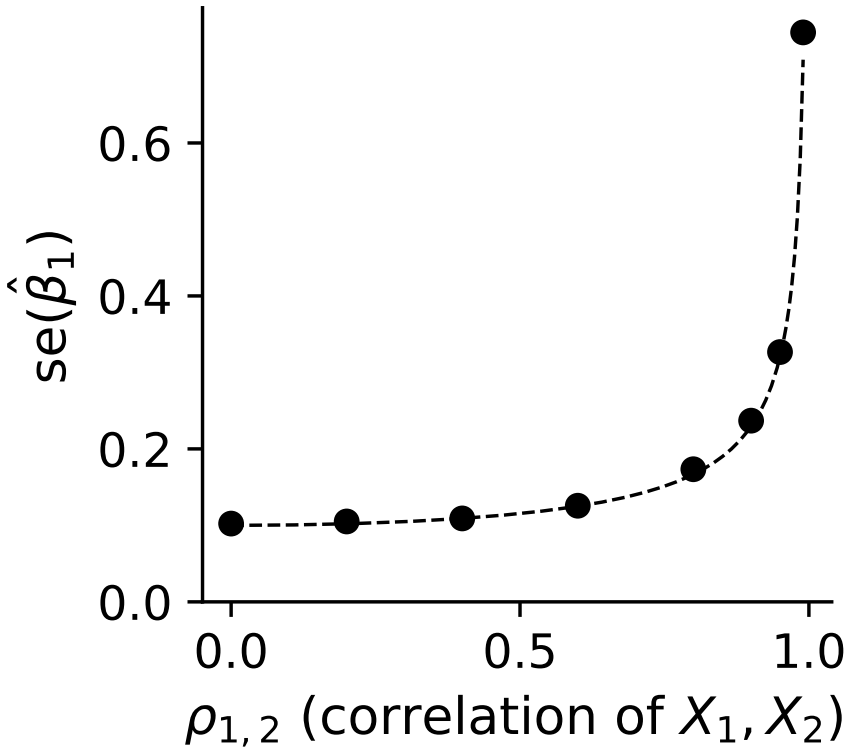




Points: 2000 simulated datasets for each $\rho_{1,2}$. Dashed: $\text{corr}(\hat{\beta}_1, \hat{\beta}_2) = -\rho_{1,2}$ and $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / \sqrt{N(1-\rho_{1,2}^2)}$.

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At $\rho_{1,2} = 0.95$, $\text{se}(\hat{\beta}_1)$ is about 3 times larger than with uncorrelated predictors, but $\hat{\beta}_1 + \hat{\beta}_2$ is as precise as ever.

Practice

For each situation, decide what changes in the model (the correlation between predictors, the spread of a predictor, or the noise) and what that does to the standard errors.

Question

1) Term limits are introduced, so half of all incumbents must leave office each election, however safe their seat. How does this change our ability to estimate the incumbency advantage?

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3) Local scandals randomly affect a few races in every election. How do they enter the model, and what do they do to the standard errors?

Where the inflation factor comes from (optional)

Two predictors: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$. Split X_1 into the part predicted by X_2 and the rest, a noise term η:

$$X_1 = a + bX_2 + \eta, \qquad \text{cov}(\eta, X_2) = 0, \qquad \sigma_\eta^2 = \sigma_{X_1}^2 \left(1 - \rho_{1,2}^2\right).$$

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With X_2 accounted for, β_1 is the slope of a single-predictor model with predictor η and noise ϵ. So the single-predictor formula, with σ_η in place of σ_{X_1}, gives

$$\operatorname{se}(\hat{\beta}_1) \approx \frac{\sigma_\epsilon}{\sigma_\eta \sqrt{N}} = \frac{\sigma_\epsilon}{\sigma_{X_1} \sqrt{N}} \cdot \frac{1}{\sqrt{1 - \rho_{1,2}^2}}.$$

Only the part of X_1 that X_2 does not share helps to estimate β_1. In the data, $R_j^2 = \hat{\rho}_{1,2}^2$ estimates $\rho_{1,2}^2$.